Thang D. Chu

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chuducthang77.github.io | Google Scholar | LinkedIn

Research Interests: reinforcement learning theory, stochastic optimization, continual learning, and LLM post-training.

Education

University of Alberta — Edmonton, AB

M.Sc. in Computing Science; GPA: 4.00/4.00 — *Jan. 2024 – Expected Aug. 2026*

Advisor: Prof. Csaba Szepesvári.

Thesis: Lower Bound on Policy Gradient with Decreasing Stepsizes.

University of Alberta — Edmonton, AB

B.Sc. in Computing Science; GPA: 3.96/4.00 — *Sep. 2019 – Apr. 2023*

Publications and Preprints

[1] **T. D. Chu**, T. Kitamura, J. Mei, C. Szepesvári, and T. Lattimore. *Lower Bound on Policy Gradient with Decreasing Stepsizes.* 2026. Working paper.

[2] T. Kitamura, A. Ghosh, A. Ayoub, **T. D. Chu**, and C. Szepesvári. *Revisiting Subgradient Dominance in Robust MDPs: Counterexamples, Hardness, and Sufficient Conditions.* 2026. Preprint. [paper]

[3] S. M. Robertson*, **T. D. Chu***, B. Dai, D. Schuurmans, C. Szepesvári, and J. Mei. *REINFORCE Converges to Optimal Policies with Any Learning Rate. Advances in Neural Information Processing Systems 38 (NeurIPS 2025).* [paper]

[4] **T. D. Chu**, T. T. Nguyen, B. D. Hai, Q. H. Nguyen, and T. Nguyen. *Graph Transformer for Drug Response Prediction. IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 20(2):1065–1072, 2023. [paper]

*Equal contribution.

Research Experience

Bilevel Soft Actor-Critic: Asymptotic and Non-Asymptotic Convergence Rates — Carleton University

Research Assistant; Advisor: Prof. Junfeng Wen — *Oct. 2022 – Jun. 2025*

- Designed a bilevel training strategy to accelerate the convergence of soft actor-critic algorithms.
- Established asymptotic convergence and derived non-asymptotic rates and optimality guarantees for the resulting method.

Kernel Representations of Gaussian Processes — University of Alberta

Research Assistant; Advisor: Prof. Martha White — *May 2022 – Sep. 2022*

- Characterized the conditions under which Gaussian processes and Bayesian linear regression induce equivalent kernel representations across different parameterizations.

Graph Transformer for Drug Response Prediction — National Economics University

Research Assistant; Advisor: Nguyen Thanh Tuan — *Jan. 2021 – Sep. 2022*

- Developed GraTransDRP by combining graph-transformer drug representations with CNN-based multi-omics encoders and KernelPCA for transcriptomic features.
- Evaluated the model on drug-response benchmarks, achieving a Pearson correlation of $r = 0.93$.

Awards

Dean's Silver Medal in Science	2023
Dean's Honor Roll	2019–2022
International Student Scholarship	2019–2022
Faculty of Science Gold Standard Scholarship	2019
University of Alberta Maple Leaf First Year Excellence Scholarship	2019

Selected Project

Large-Vessel-Occlusion Detection from EEG (JATT): Developed EEG- and clinical-data classifiers for large-vessel-occlusion detection, achieving 90% accuracy.

Skills

Programming: Python, Julia, C++, SQL, MATLAB, and R | **Machine Learning:** JAX and PyTorch